

Ema Lovrić, 28.8.2026.

Assignment 18: Virtualization solutions for system administration

<https://github.com/elovric13/eutech/tree/virtualization-solutions-for-system-administration-elovric13>

1. Explain briefly what virtualization is and its importance in computer system administration.

Virtualization is a technology that allows running multiple operating systems on a single physical machine at the same time. Instead of needing five physical servers for five different services, one server can have five virtual machines running on it. Each virtual machine thinks it has its own hardware, completely isolated from the others.

In system administration, virtualization reduces hardware costs, makes it easy to test software without risking production systems, allows fast backups and recovery through snapshots, and enables services to be separated so that if one crashes it doesn't affect everything else.

2. Analyze the following virtualization solutions: VirtualBox, Microsoft Hyper-V, Proxmox VE. For each solution, examine: Supported operating systems. Basic system requirements. Virtual machine creation and management. CPU, memory, and storage management. Snapshot capabilities. Virtual networking. Main advantages and limitations.

VirtualBox

- Supported operating systems

Runs on Windows, Linux, macOS, and Solaris as a host. Supports virtually any guest OS – Windows, Linux, BSD, Solaris, even old systems like DOS.

- Basic system requirements

Minimum 512MB RAM (2GB+ recommended for comfortable use), x86/AMD64 CPU with hardware virtualization support (Intel VT-x or AMD-V), and any modern storage. Very lightweight compared to the others.

- VM creation and management

Done through a graphical GUI that is beginner-friendly. VMs are created step by step with a wizard. Management is done from the main VirtualBox window. Also has a command-line interface (VBoxManage) for scripting and automation.

- CPU, memory, and storage management

You assign CPU cores, RAM, and disk size at VM creation. Storage uses virtual disk files (.vdi, .vmdk, .vhd formats supported). Supports dynamic storage allocation, meaning the disk file only uses actual space as data is written, not the full allocated size upfront.

- Snapshot capabilities

Full snapshot support – snapshots can be taken at any point, capturing the exact state of the VM. You can restore to any snapshot instantly. Snapshots can be branched, creating a tree of states. Very useful for testing.

- Virtual networking

Supports NAT, Bridged, Host-only, Internal Network, and NAT Network modes. NAT is the default and easiest for internet access. Bridged puts the VM on the same network as the host. Host-only creates an isolated network between VMs and the host.

- Advantages

Free and open source. Cross-platform – runs on any major OS. Beginner-friendly GUI. Excellent for desktop use, learning, and testing.

- Limitations

Performance is lower than bare-metal hypervisors. Not suitable for production server environments. GUI can feel slow with many VMs. No enterprise support unless you purchase the Extension Pack.

Microsoft Hyper-V

- Supported operating systems

Available on Windows 10/11 Pro and Enterprise, and Windows Server. As a host it requires Windows. Guest OS support is excellent for Windows systems and good for Linux (with integration services), and it supports FreeBSD.

- Basic system requirements

64-bit CPU with hardware virtualization (Intel VT-x / AMD-V) and SLAT (Second Level Address Translation). Minimum 4GB RAM for the host. Hyper-V is a Type 1 hypervisor – it runs directly on hardware, not on top of an OS like VirtualBox.

- VM creation and management

Managed through Hyper-V Manager (GUI) or PowerShell. PowerShell integration is very strong, making Hyper-V well suited for automation and scripting in Windows environments. Also manageable through Windows Admin Center.

- CPU, memory, and storage management

Supports Dynamic Memory – RAM is automatically adjusted up and down based on what the VM actually needs at any moment, rather than being fixed. Supports virtual hard disks in .vhd and .vhdx formats. VHDX supports up to 64TB virtual disk size.

- Snapshot capabilities

Called "checkpoints" in Hyper-V. Supports standard checkpoints (like traditional snapshots) and production checkpoints (which use backup technology inside the VM for application-consistent states). Very reliable for Windows Server environments.

- Virtual networking

Uses virtual switches – External (connected to physical network), Internal (host and VMs only), and Private (VMs only, no host access). Virtual switches are created and managed through Hyper-V Manager or PowerShell.

- Advantages

Type 1 hypervisor – better performance than VirtualBox. Deep Windows integration. Free with Windows Pro/Server licenses. Strong PowerShell automation. Good for Windows-heavy enterprise environments.

- Limitations

Windows host required – cannot run on Linux or macOS natively. Linux guest support is functional but not as seamless as Windows guests. Not ideal for mixed OS environments. Less beginner-friendly than VirtualBox.

Proxmox VE

- Supported operating systems

Proxmox itself runs on Debian Linux – it is installed directly on bare metal as the host OS. It supports both full VMs (KVM) running any OS, and Linux containers (LXC) for lightweight Linux environments.

- Basic system requirements

64-bit CPU with Intel VT-x or AMD-V. Minimum 2GB RAM (8GB+ recommended for production). Dedicated server hardware recommended – Proxmox is designed to be the only thing running on the machine, not installed alongside another OS.

- VM creation and management

Managed entirely through a web-based interface accessible from any browser on the network. No need to be physically at the server. Also, fully scriptable via API and command line. Supports VM templates for rapid deployment of identical machines.

- CPU, memory, and storage management

Fine-grained control over CPU cores, CPU type, memory, and storage. Supports multiple storage backends – local disk, NFS, iSCSI, Ceph distributed storage. Supports memory ballooning (dynamic RAM adjustment similar to Hyper-V Dynamic Memory). Can manage storage pools across multiple physical nodes.

- Snapshot capabilities

Full snapshot support for KVM VMs. LXC containers also support snapshots. Snapshots integrate with Proxmox Backup Server for scheduled, automated, and deduplicated backups across the entire infrastructure.

- Virtual networking

Supports Linux bridges, VLANs, bonding, and Open vSwitch. Very flexible networking – you can build complex virtual networks between VMs. Designed for multi-node cluster networking.

- Advantages

Free and open source (enterprise subscription available but optional). Type 1 hypervisor – excellent performance. Web-based management from anywhere. Supports both VMs and containers. Built-in clustering – multiple physical servers managed as one. Proxmox Backup Server integration. Best choice for homelabs and small-to-medium enterprise server environments.

- Limitations

Requires dedicated hardware – not installed alongside another OS. Steeper learning curve than VirtualBox. Linux knowledge helpful. Enterprise support requires a paid subscription.

3. Create a comparison table showing the main differences between the three virtualization solutions.

Feature	VirtualBox	Hyper-V	Proxmox VE
Type	Type 2 (hosted)	Type 1 (bare-metal)	Type 1 (bare-metal)
Host OS	Windows, Linux, macOS	Windows only	Debian Linux (bare metal)
Guest OS support	Any OS	Windows, Linux	Any OS (KVM) + Linux containers
Cost	Free	Free with Windows	Free (paid support optional)
Management interface	GUI + CLI	Hyper-V Manager + PowerShell	Web browser
Dynamic memory	No	Yes	Yes
Snapshots	Yes	Yes	Yes
Clustering	No	Yes (with Windows Server)	Yes (built-in)
Performance	Lower (hosted)	High	High
Best for	Desktop testing, learning	Windows enterprise	Server management, homelabs
Learning curve	Low	Medium	Medium-high

4. Determine which solution would be most appropriate for:

- Managing multiple virtual machines from a server.

Proxmox VE – it is designed exactly for this. The web interface lets manage dozens of VMs across one or multiple physical servers from any browser, with storage pools, backups, and clustering built in.

- Creating environments for operating system testing.

VirtualBox – the easiest to set up on a personal workstation, supports any guest OS, has excellent snapshot support for reverting after tests, and costs nothing. No need for a dedicated server just to test an OS.

- Keeping different services separated through virtual machines.

Proxmox VE – service isolation is one of its core use cases. Running a web server, database, and mail server in separate VMs on one physical machine, all isolated from each other, is exactly what Proxmox is built for.

5. Select one of the three solutions as your main recommendation and explain the reasons for your choice.

Proxmox VE – the most complete and most realistic tool. It mirrors what you encounter in real enterprise and cloud environments – Type 1 hypervisor, web-based management, VM and container support, snapshots, backups, and clustering. It is free, actively developed, and has a large community. VirtualBox is excellent for learning on a personal laptop but is not what production systems use. Hyper-V is strong but locks you into Windows. Proxmox works with any OS and teaches the skills that transfer directly to professional environments.

6. Explain at least five benefits of using virtualization in computer system administration.

1. Hardware cost reduction: Multiple services run on one physical machine instead of requiring separate hardware for each. A server that would previously host one service now hosts ten, dramatically reducing hardware, power, and space costs.
2. Fast disaster recovery: Snapshots allow the entire state of a VM to be saved and restored in seconds. If a software update breaks a system, rolling back to a pre-update snapshot takes less time than any manual recovery process.
3. Service isolation: Each service runs in its own VM. If one VM crashes or is compromised, it does not affect the others. A security breach in one VM does not automatically spread to the entire physical machine.
4. Easy testing and development: Developers and administrators can create a complete test environment, experiment freely, and destroy it when done – without touching production systems. This removes fear from testing and encourages experimentation.
5. Simplified backups and migration: A VM is ultimately just a set of files. Backing up a VM means copying those files. Migrating a VM to different hardware means moving those files. This is far simpler than managing physical machine migrations.

7. Identify three potential problems or limitations associated with virtual machines and explain how they can be reduced or managed.

- Single point of hardware failure

If the physical host machine fails, all VMs on it go down simultaneously – more impact than if only one physical server failed.

How to manage it: Use of clustering (Proxmox, Hyper-V) so VMs automatically migrate to another physical node if one fails. Maintaining regular backups on separate storage.

- Resource contention

Multiple VMs share the same physical CPU, RAM, and disk. If one VM suddenly consumes all resources (a "noisy neighbour"), other VMs suffer performance degradation.

How to manage it: Setting resource limits and reservations per VM. Use of monitoring tools to detect abnormal resource usage early and acting before it impacts other VMs.

- Snapshot sprawl

Snapshots are easy to create and easy to forget. Over time, hundreds of snapshots accumulate, consuming large amounts of storage and slowing down VM performance.

How to manage it: Establishing a snapshot policy – limiting the number of snapshots per VM, setting retention periods, and use of automated backup tools instead of manual snapshots for long-term recovery points.

8. References

Oracle VirtualBox Documentation - <https://docs.oracle.com/en/virtualization/virtualbox/>

Microsoft Hyper-V Documentation - <https://learn.microsoft.com/en-us/virtualization/hyper-v-on-windows/>

Proxmox VE Documentation - https://pve.proxmox.com/wiki/Main_Page

Red Hat - <https://www.redhat.com/en/topics/virtualization/what-is-a-hypervisor>

VMware - <https://www.vmware.com/topics/hypervisor#types-of-hypervisors>